

Lecture 2 : Calculus and Linear Algebra

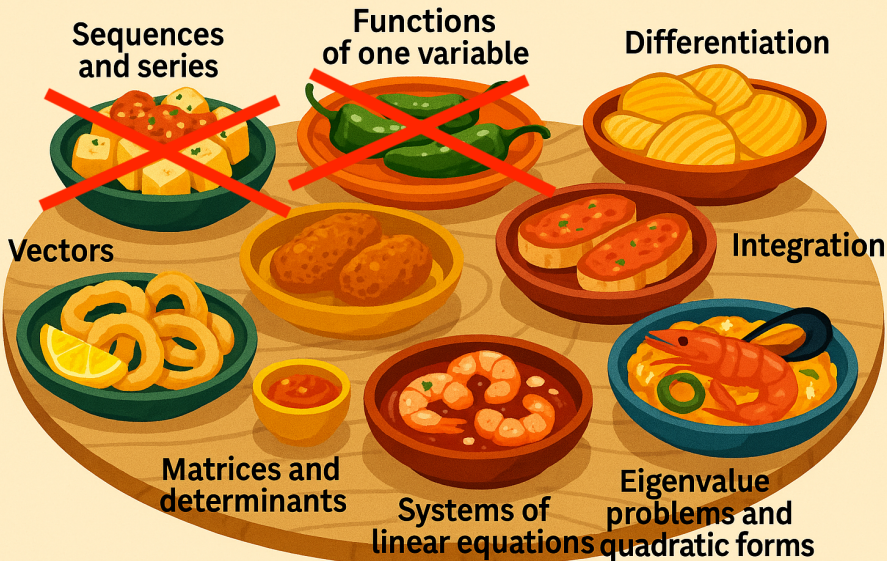
Piotr Zwiernik

Mathematics Brush-up



Barcelona School of Economics

Course content (math tapas)



Optimization in 1D

Monotonicity via derivatives

If f is differentiable on (a, b) :

1. f increasing on $[a, b] \iff f'(x) \geq 0$ on (a, b) .
2. f decreasing on $[a, b] \iff f'(x) \leq 0$ on (a, b) .
3. f constant on $[a, b] \iff f'(x) = 0$ on (a, b) .
4. $f'(x) > 0$ on $(a, b) \Rightarrow f$ strictly increasing on $[a, b]$.
5. $f'(x) < 0$ on $(a, b) \Rightarrow f$ strictly decreasing on $[a, b]$.

Converse of 4-5 can fail at isolated points (e.g., x^3).

Example: $f(x) = \frac{x^3}{3} + 2x^2 + 3x + 1$; $f'(x) = (x+1)(x+3)$ determines monotonicity intervals and local extrema.

Optimal points: first-order conditions

Local optima

Let $f : D \rightarrow \mathbb{R}$. A point $x_0 \in D$ is a **local maximum** if $\exists (a, b) \subset D$ with $x_0 \in (a, b)$ and $f(x) \leq f(x_0)$ for all $x \in (a, b)$. Similarly for **local min** with $\geq$.

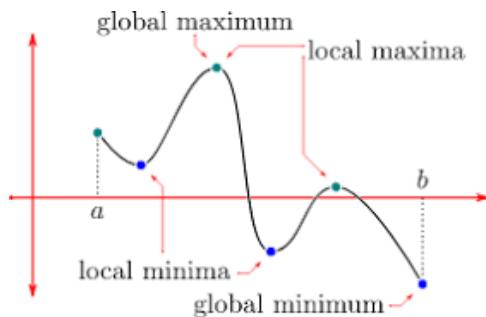
If f has a local optimum at interior x and is differentiable, then $f'(x) = 0$.

To see why this is true, stare at $f(x_0 + h) - f(x_0) \approx f'(x_0)h$.

This condition is **necessary**, not sufficient; see e.g. x^3 .

If x is stationary ($f'(x) = 0$) and f is strictly increasing to the left and strictly decreasing to the right, then x is a local max (reverse for min).

Optimal points



Example: Timing a sale (present value)

Suppose an assets market value is $V(t)$ at time t . At a constant annual interest rate R , the **present value** of selling at time t is

$$P(t) = V(t) e^{-Rt},$$

where e^{-Rt} discounts future cash flows.

Goal: Choose t to maximize $P(t)$.

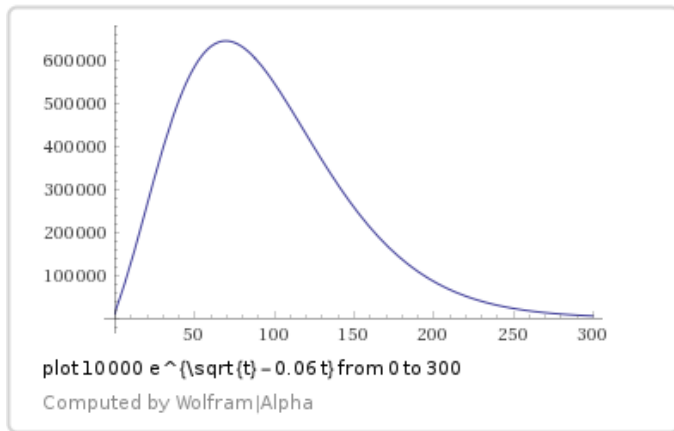
FOC (set derivative to zero):

$$P'(t) = V'(t)e^{-Rt} - R V(t)e^{-Rt} = 0 \quad \Rightarrow \quad \frac{V'(t)}{V(t)} = R.$$

Interpretation: Sell when the assets *instantaneous percentage growth rate* (matches) the *interest rate you could earn in the bank*.

Example: If $V(t) = 10000 e^{\sqrt{t}}$ and $R = 6\%$, then $P(t) = 10000 e^{\sqrt{t} - 0.06 t}$, which is maximized near $t \approx 69.44$.

Plot of the present value $P(t)$



To certify a **global** optimum on an interval, compare values at boundaries and critical points. Here $P(0) = 10000$, $\lim_{t \rightarrow \infty} P(t) = 0$.

Second-order conditions

Test: If $f'(x_0) = 0$ and $f''(x_0) < 0$ (> 0), then x_0 is a strict local max (min).

Indeed,
$$f(x_0 + h) \approx f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 = f(x_0) + \frac{1}{2}f''(x_0)h^2.$$

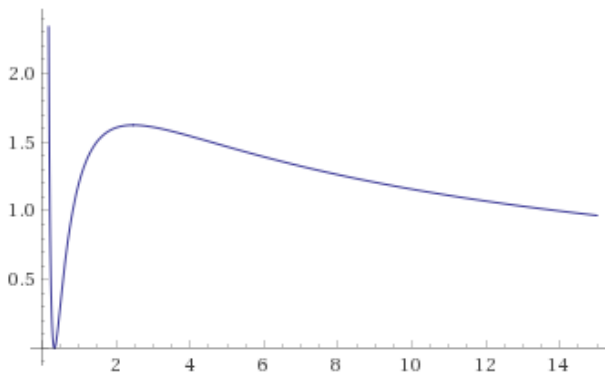
Example: For $f(x) = \frac{\log^2(3x)}{x}$ ($x > 0$):

$$f'(x) = \frac{(2 - \log 3x) \log 3x}{x^2}, \quad f''(x) = \frac{2(1 - 3 \log 3x + \log^2 3x)}{x^3}.$$

Critical points at $x_1 = \frac{1}{3}$ (min), $x_2 = \frac{e^2}{3}$ (max).

Question: What if $f''(x_0) = 0$? (assume all derivatives exist)

Plot of $f(x) = \frac{\log^2 3x}{x}$, $x > 0$



plot $\ln^2(3x)/x$ from 0 to 15 | Computed by Wolfram|Alpha

$\lim_{x \rightarrow \infty} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = \infty$, $f(x_1) = 0$, $f(x_2) > 0 \Rightarrow x_1$ global min,
no global max.

Convexity and concavity

If f is twice differentiable on (a, b) :

1. f convex on $[a, b] \Leftrightarrow f''(x) \geq 0$ for all $x \in (a, b)$.
2. f concave on $[a, b] \Leftrightarrow f''(x) \leq 0$ for all $x \in (a, b)$.
3. $f'' > 0$ (< 0) $\Rightarrow f$ strictly convex (concave).

Theorem: If f is convex and nondecreasing, and g is convex, then $f \circ g$ is convex. If f is convex and nonincreasing, and g is concave, then $f \circ g$ is convex.

Example: $f(x) = e^{x^2}$ is strictly convex on $[0, \infty)$.

Economics:

- **Cost functions:** usually convex — producing extra units becomes progressively more expensive (e.g. overtime wages, machine wear).
- **Utility functions:** often concave — reflects diminishing marginal utility (the first slice of pizza gives more satisfaction than the tenth).
- **Machine learning:** convex losses (e.g. squared error, logistic loss + ℓ_2 regularization) ensure optimization problems are tractable and stable.

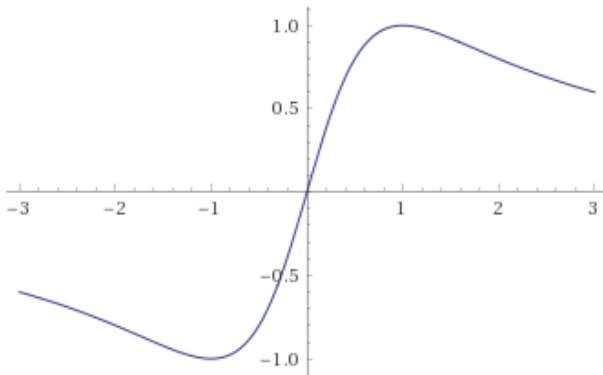
Example of changing curvature

Let $f(x) = \frac{2x}{x^2 + 1}$. Then

$$f'(x) = \frac{2(1 - x^2)}{(x^2 + 1)^2}, \quad f''(x) = \frac{4(x^3 - 3x)}{(x^2 + 1)^3}.$$

So f is strictly convex on $[-\sqrt{3}, 0] \cup [\sqrt{3}, \infty)$ and strictly concave on $(-\infty, -\sqrt{3}] \cup [0, \sqrt{3}]$.

Plot of $f(x) = \frac{2x}{x^2 + 1}$



plot $2x/(x^2+1)$ from -3 to 3 | Computed by Wolfram|Alpha

Inflexion points at $x \in \{-\sqrt{3}, 0, \sqrt{3}\}$. Global max at $x = 1$, min at $x = -1$, and $\lim_{|x| \rightarrow \infty} f(x) = 0$.

Inflexion points

If f is C^n on (a, b) , an **inflexion point** at x_0 occurs when curvature changes sign. A sufficient test:

$$f''(x_0) = \dots = f^{(n-1)}(x_0) = 0, \quad f^{(n)}(x_0) \neq 0, \quad n \text{ odd.}$$

Example: $f(x) = x^3$ has $f''(0) = 0$, $f^{(3)}(0) = 6 \neq 0 \Rightarrow$ inflexion at 0.

Taylor series expansion

Mean-value theorem

Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad (\text{equivalently } f(b) = f(a) + f'(c)(b - a)).$$

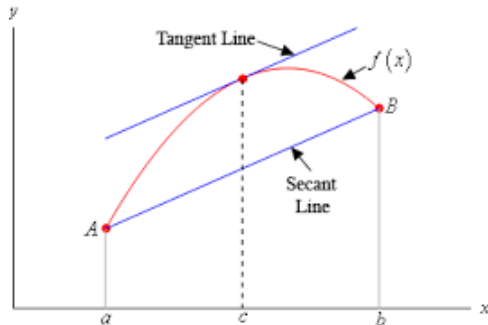
Proof idea: consider $h(x) = (f(b) - f(a))x - (b - a)f(x)$; since $h(a) = h(b)$, h has an interior extremum.

Interpretation: the instantaneous slope matches the secant slope at some point.

Geometry of the mean-value theorem

Recall: $\exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Taylor formula

Theorem (Taylor's formula of order n): Let f be $n + 1$ times differentiable on (a, b) , and let $x_0 \in (a, b)$ given. Then, for any $x \in (a, b)$, if $n = 1$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \text{error}.$$

If $n \geq 2$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \text{error}.$$

The error is small if x is close to x_0 , therefore, this theorem says that differentiable functions may be **locally approximated by a polynomial** called the **Taylor polynomial**.

Taylor remainder and a handy expansion

Remainder (Lagrange form)

$$R_n^y(x) = \frac{f^{(n+1)}(y)}{(n+1)!} (x - x_0)^{n+1} \quad \text{for some } y \text{ between } x_0 \text{ and } x.$$

If $|f^{(n+1)}| \leq M$ near x_0 , then $|R_n^y(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$.

Example (log around 1): With $x_0 = 1$,

$$\log x \approx (x - 1) - \frac{1}{2}(x - 1)^2, \quad R_2^y(x) = \frac{1}{3y^3}(x - 1)^3.$$

Short comment about small-o notation

Writing $|R_n^y(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$ we conclude

$$\frac{R_n^y(x)}{|x - x_0|^n} \leq \frac{M}{(n+1)!} |x - x_0| \text{ and so}$$

$$\lim_{x \rightarrow x_0} \frac{R_n^y(x)}{|x - x_0|^n} = 0.$$

This we write as $R_n^y(x) = o(|x - x_0|^n)$.

In particular: If f is twice differentiable:

$$f(x+h) = f(x) + f'(x)h + o(|h|).$$

The generalization of this last expression to multivariate functions will provide powerful insights into their optimization.

Chapter 4: Integration

Often a function f is given and we are looking for F whose derivative is f .

Example: the marginal cost function $C'(x)$ is known (how cost changes with production x). We want the cost $C(x)$ itself.

Idea: **integration** reverses differentiation: it accumulates small changes.

Read Chapter 5 of Werner-Sotskov **Exercises** 5.1(a)-(b), 5.2(a)-(b), 5.3(c)

Indefinite integrals

Indefinite integrals

A differentiable F is an **antiderivative** of f if $F'(x) = f(x)$ on a common domain.

Fact: all antiderivatives differ by a constant: if $F'(x) = f(x)$, then any $\tilde{F}(x) = F(x) + c$ also satisfies $\tilde{F}'(x) = f(x)$.

Definition: the **indefinite integral** is

$$\int f(x) dx = F(x) + c.$$

Linearity:

1. $\int (f + g) dx = \int f dx + \int g dx$
2. $\int c f dx = c \int f dx$

Indefinite integrals you should know

Templates:

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$$

$$2. \int \frac{1}{x} dx = \log |x| + c$$

$$3. \int e^x dx = e^x + c$$

$$4. \int \sin x dx = -\cos x + c$$

$$5. \int \cos x dx = \sin x + c$$

$$6. \int a^x dx = \frac{a^x}{\log a} + c \quad (a > 0, a \neq 1)$$

Always check by differentiating the right-hand side.

Integration by substitution

Theorem (Substitution)

If $t = g(x)$ and F is an antiderivative of f , then

$$\int f(g(x)) g'(x) dx = \int f(t) dt = F(t) + c = F(g(x)) + c.$$

(if $F'(t) = f(t)$ then $f(g(x))g'(x) = (F(g(x)))'$ by the chain rule)

Examples:

$$1. \int (ax + b)^n dx = \frac{1}{a} \int t^n dt = \frac{(ax + b)^{n+1}}{a(n+1)} + c.$$

($t = ax + b$, so $dt = a dx$.)

$$2. \int \frac{e^x}{\sqrt[3]{1 + e^x}} dx = \int t^{-1/3} dt = \frac{3}{2} t^{2/3} + c = \frac{3}{2} (1 + e^x)^{2/3} + c.$$

($t = 1 + e^x$, so $dt = e^x dx$.)

Integration by parts

Theorem (Integration by parts)

For differentiable u, v ,

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx.$$

Proof: Differentiate $u(x)v(x)$ and integrate.

Example: with $u = \log x$, $v' = 1$,

$$\int \log x dx = x \log x - \int 1 dx = x(\log x - 1) + c.$$

A combo: substitution then parts

Compute $\int \sin \sqrt{x} \, dx$.

Substitute $t = \sqrt{x}$, so $x = t^2$ and $dx = 2t \, dt$:

$$\int \sin \sqrt{x} \, dx = 2 \int t \sin t \, dt.$$

Now parts with $u = t$, $v' = \sin t$:

$$2 \int t \sin t \, dt = 2 \left(-t \cos t + \int \cos t \, dt \right) = 2(-t \cos t + \sin t) + c.$$

Back-substitute $t = \sqrt{x}$:

$$\boxed{\int \sin \sqrt{x} \, dx = 2 \left(-\sqrt{x} \cos \sqrt{x} + \sin \sqrt{x} \right) + c}.$$

A combo: substitution then parts



Definite integrals

Definite integral as area and accumulation

For continuous $f : [a, b] \rightarrow \mathbb{R}$ with $f \geq 0$, the definite integral

$$\int_a^b f(x) dx$$

is the area under f between a and b . It accumulates signed change.

Properties:

1. $\int_b^a f = -\int_a^b f$
2. $\int_a^b cf = c \int_a^b f$
3. If $c \in [a, b]$, then $\int_a^b f = \int_a^c f + \int_c^b f$
4. $\left| \int_a^b f \right| \leq \int_a^b |f|$
5. If $f \leq g$ on $[a, b]$, then $\int_a^b f \leq \int_a^b g$

Some uses: cumulative revenue from a known marginal revenue curve, energy used by a device with power draw $P(t)$, or probability mass from a density.

Fundamental Theorem of Calculus

Fundamental Theorem of Calculus

If f is continuous on $[a, b]$ and F is an antiderivative of f , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Moreover, $G(t) = \int_a^t f(x) dx$ is differentiable and $G'(t) = f(t)$.

Marginal to total: If $C'(x)$ is marginal cost, then the change in total cost from $x = 300$ to $x = 400$ is

$$C(400) - C(300) = \int_{300}^{400} C'(x) dx.$$

Example: $C'(x) = 6 - \frac{60}{x+1}$ for $x \in [0, 1000]$:

$$\int_{300}^{400} \left(6 - \frac{60}{x+1}\right) dx = \left(6x - 60 \log |x+1|\right)_{300}^{400} \approx 582.79.$$

Application: proving $e = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$ via integrals

Define $\log x = \int_1^x \frac{1}{t} dt$. Let e^x be the inverse of $\log x$; then $1 = \int_1^e \frac{1}{t} dt$.

For $t \in [1, 1 + \frac{1}{n}]$,

$$\frac{1}{n+1} = \int_1^{1+\frac{1}{n}} \frac{1}{1+\frac{1}{n}} dt \leq \int_1^{1+\frac{1}{n}} \frac{1}{t} dt \leq \int_1^{1+\frac{1}{n}} 1 dt = \frac{1}{n}.$$

So

$$\frac{1}{n+1} \leq \log \left(1 + \frac{1}{n} \right) \leq \frac{1}{n}.$$

Exponentiate and rearrange to obtain

$$\frac{e}{1 + \frac{1}{n}} \leq \left(1 + \frac{1}{n} \right)^n \leq e,$$

and let $n \rightarrow \infty$.



Application: Taylor with integral remainder (up to $n = 2$)

Start with Fundamental Theorem of Calculus (FTC):

$$f(x) = f(x_0) + \int_{x_0}^x f'(t_1) dt_1.$$

Apply FTC again to $f'(t_1)$:

$$f'(t_1) = f'(x_0) + \int_{x_0}^{t_1} f''(t_2) dt_2.$$

Substitute:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \int_{x_0}^x \int_{x_0}^{t_1} f''(t_2) dt_2 dt_1.$$

Repeating once more gives the quadratic term plus an integral remainder:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + R_2(x).$$

Chapter 5: Vectors

A **vector** is an ordered n -tuple of real numbers: $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$.

Why care in econ/data:

- a bundle of n goods (quantities),
- a user's features or a product's attributes,
- a portfolio's weights across n assets,
- a document's word counts or an embedding.

Read Chapter 6 of Werner-Sotskov; Simon-Blume Chs. 10-11.

Exercises 6.2, 6.3, 6.4, 6.6, 6.7, 6.8

Vectors and the Euclidean space

Definition and notation

A **vector** $\mathbf{v}$ is an ordered n -tuple $(v_1, \dots, v_n)$ of real numbers called **coordinates**.

Notation: $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$.

The zero vector is $\mathbf{0} = (0, \dots, 0)$.

The i -th unit vector is $\mathbf{e}_i$ (a 1 in position i , zeros elsewhere).

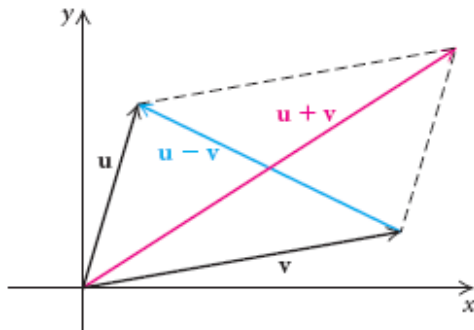
Operations:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \dots, u_n + v_n), \quad \lambda \mathbf{v} = (\lambda v_1, \dots, \lambda v_n).$$

These satisfy the usual commutative, associative, and distributive laws.

Sum and difference of two vectors

Note $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-1)\mathbf{v}$.



Inner product and norm

The **inner product** (dot product) of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i v_i.$$

The **Euclidean norm** is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

Properties:

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$
3. **Cauchy-Schwarz:** $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$
4. **Triangle inequality:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

Example (platform revenue): hours watched per genre $\mathbf{u} = (500, 200, 50)$ and euro-per-hour rates $\mathbf{v} = (2, 3, 5)$ yield total revenue $\langle \mathbf{u}, \mathbf{v} \rangle = 2150$.

The Law of Cosines and angles

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\| \|\mathbf{v}\| \cos(\angle(\mathbf{u}, \mathbf{v})).$$

Equivalently,

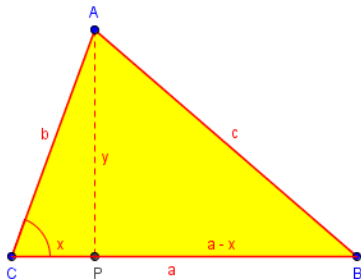
$$\cos(\angle(\mathbf{u}, \mathbf{v})) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

We prove that $c^2 = a^2 + b^2 - 2ab \cos(C)$:

By Pythagoras, $x^2 + y^2 = b^2$ and $(a-x)^2 + y^2 = c^2$.

Subtract to eliminate y^2 to get $c^2 = a^2 + b^2 - 2ax$.

Use $\cos C = x/b$.



Orthogonality and linear independence

Orthogonality

Definition: $\mathbf{u} \perp \mathbf{v}$ if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Angle example: For $\mathbf{u} = (2, -1, 3)$ and $\mathbf{v} = (5, -4, -1)$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = 11, \quad \cos \angle(\mathbf{u}, \mathbf{v}) = \frac{11}{\sqrt{14}\sqrt{42}} \approx 0.4537,$$

so the angle is about 63° .

Geometric test: $\mathbf{u} \perp \mathbf{v}$ if and only if $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Cauchy-Schwarz via completing the square

Assume nonzero $\mathbf{u}, \mathbf{v}$. For any real t ,

$$0 \leq \|\mathbf{u} - t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2.$$

Complete the square in t :

$$\|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2 = \|\mathbf{v}\|^2 \left(t - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \right)^2 + \left(\|\mathbf{u}\|^2 - \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{v}\|^2} \right).$$

Since the left side is ≥ 0 for all t , the second term must be ≥ 0 :

$$\langle \mathbf{u}, \mathbf{v} \rangle^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2.$$

Equality condition: equality holds if and only if $\|\mathbf{u} - t\mathbf{v}\|^2 = 0$ for $t = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$, i.e., $\mathbf{u} = t\mathbf{v}$ (collinear vectors).



Linear dependence, independence, and bases

Linear combination: $\mathbf{u} = \sum_{i=1}^m \lambda_i \mathbf{v}_i$.

Linear independence: $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent if

$$\sum_{i=1}^m \lambda_i \mathbf{v}_i = \mathbf{0} \quad \Rightarrow \quad \lambda_1 = \dots = \lambda_m = 0.$$

Basis: Any n linearly independent vectors in $\mathbb{R}^n$ form a basis. Then every $\mathbf{u} \in \mathbb{R}^n$ has a **unique** representation $\mathbf{u} = \sum_{i=1}^n \lambda_i \mathbf{v}_i$.

Standard basis: $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.